Observability Evaluation in Map-Assisted Radio SLAM

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***Abstract*—**

I. Introduction

II. Theoretical Background: Observability Measures

The observability of a dynamic system determines whether the system state can be uniquely determined from a finite set of observations. This section provides the mathematical foundation for evaluating the localization model using map information in radio SLAM system.

III. Model Description

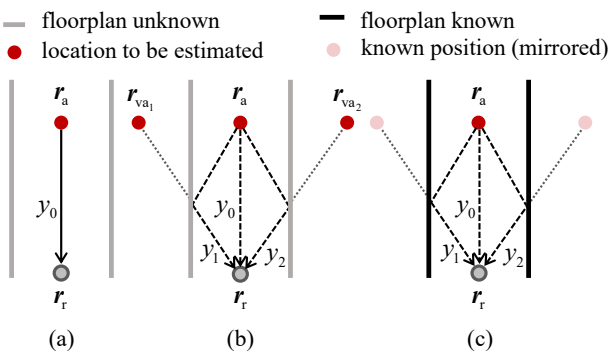

Fig. 1. Caption

A. Dynamics Model

1) Mapless model: The receiver (r) is assumed to follow a velocity random-walk model at position $\boldsymbol{r}_{\mathrm{r}}$, while the anchor (a) is modeled as a stationary transmitter at position $\boldsymbol{r}_{\mathrm{a}}$. $\boldsymbol{r}_{\mathrm{va}_i}^T$ is the position of virtual anchor generated by the i-th reflected surface, with the same clock bias related to the anchor. As shown in Figure 1(b), $\boldsymbol{r}_{\mathrm{va}_1}, \boldsymbol{r}_{\mathrm{va}_2}, \boldsymbol{r}_{\mathrm{a}}$ are the anchors (or virtual anchors) need to be estimated. The positions of the receiver and anchor incorporate a two-state clock model consisting of clock bias δt and clock drift $\dot{\delta t}$. By discretizing the augmented continuous-time dynamics with a sampling period $T \triangleq t_{k+1} - t_k$, the discrete-time model is expressed as:

$$\boldsymbol{x}(t_{k+1}) = \mathbf{F}\boldsymbol{x}(t_k) + \boldsymbol{w}(t_k), \quad k = 0, 1, 2, \ldots \tag{1}$$

where the augmented state vector $\boldsymbol{x} \triangleq [\boldsymbol{x}_{\mathrm{r}}^T, \boldsymbol{x}_{\mathrm{a}}^T, \boldsymbol{r}_{\mathrm{va}_1}^T, \cdots, \boldsymbol{r}_{\mathrm{va}_N}^T]^T$ indicates the status from the receiver, anchor and N virtual anchors. The receiver state is defined as $\boldsymbol{x}_{\mathrm{r}} = [\boldsymbol{r}_{\mathrm{r}}^T, \dot{\boldsymbol{r}}_{\mathrm{r}}^T, \delta t_{\mathrm{r}}, \dot{\delta t}_{\mathrm{r}}]^T$ and the anchor state as $\boldsymbol{x}_{\mathrm{a}} = [\boldsymbol{r}_{\mathrm{a}}^T, \delta t_{\mathrm{a}}, \dot{\delta t}_{\mathrm{a}}]^T$, where $\boldsymbol{x}_{\mathrm{a}} = [x_a, y_a]^T, \dot{\boldsymbol{x}}_{\mathrm{a}} = [\dot{x}_a, \dot{y}_a]^T, \boldsymbol{x}_{\mathrm{r}} = [x_r, y_r]^T, \dot{\boldsymbol{x}}_{\mathrm{r}} = [\dot{x}_r, \dot{y}_r]^T$.

The state transition matrix is $\mathbf{F} = \mathrm{diag}[\mathbf{F}_{\mathrm{r}}, \mathbf{F}_{\mathrm{a}}, \mathbf{F}_{\mathrm{va}}]$, with sub-matrices:

$$\mathbf{F}_{\mathrm{r}} = \begin{bmatrix} \mathbf{I}_{2\times2} & T\mathbf{I}_{2\times2} & \mathbf{0}_{2\times2} \\ \mathbf{0}_{2\times2} & \mathbf{I}_{2\times2} & \mathbf{0}_{2\times2} \\ \mathbf{0}_{2\times2} & \mathbf{0}_{2\times2} & \mathbf{F}_{\mathrm{clk}} \end{bmatrix}, \mathbf{F}_{\mathrm{clk}} = \begin{bmatrix} 1 & T \\ 0 & 1 \end{bmatrix}, \tag{2}$$

$$\mathbf{F}_{\mathrm{a}} = \mathrm{diag}[\mathbf{I}_{2\times2}, \mathbf{F}_{\mathrm{clk}}], \mathbf{F}_{\mathrm{va}} = \mathrm{diag}[\mathbf{I}_{2N\times2N}] \tag{3}$$

The process noise $\boldsymbol{w}(t_k)$ is a zero-mean white sequence with covariance $\mathbf{Q} = \mathrm{diag}[\mathbf{Q}_{\mathrm{r}}, \mathbf{Q}_{\mathrm{a}}, \mathbf{Q}_{\mathrm{va}}]$. The blocks $\mathbf{Q}_{\mathrm{r}} = \mathrm{diag}[\mathbf{Q}_{\mathrm{pv}}, \mathbf{Q}_{\mathrm{clk,r}}]$, $\mathbf{Q}_{\mathrm{a}} = \mathrm{diag}[\mathbf{0}_{2\times2}, \mathbf{Q}_{\mathrm{clk,a}}]$, and $\mathbf{Q}_{\mathrm{va}} = \mathrm{diag}[\mathbf{0}_{2i\times2i}]$ encapsulate the velocity random-walk and clock noise intensities with:

$$\mathbf{Q}_{\mathrm{pv}} = \begin{bmatrix} \tilde{q}_x \frac{T^3}{3} & 0 & \tilde{q}_x \frac{T^2}{2} & 0 \\ 0 & \tilde{q}_y \frac{T^3}{3} & 0 & \tilde{q}_y \frac{T^2}{2} \\ \tilde{q}_x \frac{T^2}{2} & 0 & \tilde{q}_x T & 0 \\ 0 & \tilde{q}_y \frac{T^2}{2} & 0 & \tilde{q}_y T \end{bmatrix}, \tag{4}$$

$$\mathbf{Q}_{\mathrm{clk}} = \begin{bmatrix} S_{\tilde{w}_{\delta t}} T + S_{\tilde{w}_{\dot{\delta} t}} \frac{T^3}{3} & S_{\tilde{w}_{\dot{\delta} t}} \frac{T^2}{2} \\ S_{\tilde{w}_{\dot{\delta} t}} \frac{T^2}{2} & S_{\tilde{w}_{\dot{\delta} t}} T \end{bmatrix} \tag{5}$$

The parameters $\tilde{q}_x$ and $\tilde{q}_y$ denote the power spectral densities (PSD) of the zero-mean white noise processes driving the receiver's acceleration in the x and y directions, respectively. The power spectra $S_{\tilde{w}_{\delta t}}$ and $S_{\tilde{w}_{\dot{\delta} t}}$ are determined by the oscillator's white frequency coefficient h_0 and random-walk coefficient h_{-2}, which lead to $S_{\tilde{w}_{\delta t}} \approx h_0/2$, and $S_{\tilde{w}_{\dot{\delta} t}} \approx 2\pi^2 h_{-2}$ [1, 2].

2) Map-assisted model: Compared to the dynamic equations without a floorplan, adding a floorplan provides prior constraints on the mirrored positions of the anchor and virtual anchor. Ideally, if all signal reflections originate from walls, the virtual anchor position can be calculated by mirroring the anchor and floorplan positions. In Figure 1(c), $\boldsymbol{r}_{\mathrm{a}}$ is the only anchor location need to be estimated, the two virtual anchors are considered as mirrored known positions. In this case, the dynamic equation is defined as

$$\boldsymbol{x}_{\mathrm{M}}(t_{k+1}) = \mathbf{F}_{\mathrm{M}}\boldsymbol{x}_{\mathrm{M}}(t_k) + \boldsymbol{w}_{\mathrm{M}}(t_k), \quad k = 0, 1, 2, \ldots \tag{6}$$

Here, the state of the receiver is defined as $\boldsymbol{x}_{\mathrm{M}} \triangleq [\boldsymbol{x}_{\mathrm{r}}^T, \boldsymbol{x}_{\mathrm{a}}^T]^T$, the state transition matrix is defined as $\mathbf{F}_{\mathrm{M}} = \mathrm{diag}[\mathbf{F}_{\mathrm{r}}, \mathbf{F}_{\mathrm{a}}]$, and the process noise $\boldsymbol{w}_{\mathrm{M}}(t_k)$ is a zero-mean white sequence with covariance $\mathbf{Q}_{\mathrm{M}} = \mathrm{diag}[\mathbf{Q}_{\mathrm{r}}, \mathbf{Q}_{\mathrm{a}}]$.

B. Observation Model

To unify the observation models for the mapless model and map-assisted model, we formulate the range measurements as a stacked vector $\boldsymbol{z}(t_k) \in \mathbb{R}^{N+1}$, containing one direct path measurement and N reflected path measurements.

1) Mapless model: For the mapless model, the observation vector $\boldsymbol{z}(t_k) = \boldsymbol{h}(\boldsymbol{x}(t_k)) + \boldsymbol{v}(t_k)$ is given by:

$$\begin{aligned} z(t_k) &= h(\boldsymbol{x}(t_k)) + v(t_k) \\ &= ||\boldsymbol{r}_{\mathrm{r}}(t_k) - \boldsymbol{r}_{\mathrm{a}}(t_k)||_2 + c \cdot \Delta t(t_k) + v(t_k) \end{aligned} \tag{7}$$

where $\Delta t(t_k) \triangleq \delta t_{\mathrm{r}}(t_k) - \delta t_{\mathrm{a}}(t_k)$ is the relative clock bias.

To perform observability and Fisher Information analysis, the nonlinear observation is linearized via a Taylor series expansion about the nominal state. The linearized observation matrix (Jacobian) $\mathbf{H}(t_k)$ is defined as:

$$\mathbf{H}(t_k) \triangleq \left.\frac{\partial h}{\partial \boldsymbol{x}}\right|_{\hat{\boldsymbol{x}}(t_k)} = [\mathbf{H}_{\mathrm{r}}(t_k), \mathbf{H}_{\mathrm{a}}(t_k), \mathbf{H}_{\mathrm{va}}(t_k)], \tag{8}$$

where

$$\mathbf{H}_{\mathrm{r}}(t_k) = \begin{bmatrix} \hat{\boldsymbol{e}}^T(\boldsymbol{r}_{\mathrm{a}}) & \mathbf{0}_{1\times 2} & c & 0 \\ \hat{\boldsymbol{e}}^T(\boldsymbol{r}_{\mathrm{va}_1}) & \mathbf{0}_{1\times 2} & c & 0 \\ \vdots & \vdots & \vdots & \vdots \\ \hat{\boldsymbol{e}}^T(\boldsymbol{r}_{\mathrm{va}_N}) & \mathbf{0}_{1\times 2} & c & 0 \end{bmatrix}, \mathbf{H}_{\mathrm{a}}(t_k) = \begin{bmatrix} -\hat{\boldsymbol{e}}^T(\boldsymbol{r}_{\mathrm{a}}) & -c & 0 \\ \mathbf{0}_{1\times 2} & -c & 0 \\ \vdots & \vdots & \vdots \\ \mathbf{0}_{1\times 2} & -c & 0 \end{bmatrix}$$

$$\mathbf{H}_{\mathrm{va}}(t_k) = \begin{bmatrix} \mathbf{0}_{1\times 2} & \mathbf{0}_{1\times 2} & \dots & \mathbf{0}_{1\times 2} \\ -\hat{\boldsymbol{e}}^T(\boldsymbol{r}_{\mathrm{va}_1}) & \mathbf{0}_{1\times 2} & \dots & \mathbf{0}_{1\times 2} \\ \mathbf{0}_{1\times 2} & -\hat{\boldsymbol{e}}^T(\boldsymbol{r}_{\mathrm{va}_2}) & \dots & \mathbf{0}_{1\times 2} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{0}_{1\times 2} & \mathbf{0}_{1\times 2} & \dots & -\hat{\boldsymbol{e}}^T(\boldsymbol{r}_{\mathrm{va}_N}) \end{bmatrix}$$

$\hat{\boldsymbol{e}}$ is the unit direction vector from a generic position $\mathbf{p}$ (either the anchor or a virtual anchor) to the receiver as:

$$\hat{\boldsymbol{e}}(\boldsymbol{p}) \triangleq \frac{\boldsymbol{r}_{\mathrm{r}}(t_k) - \boldsymbol{p}}{||\boldsymbol{r}_{\mathrm{r}}(t_k) - \boldsymbol{p}||_2}$$

2) Map-assisted model: For the map-assisted model, the virtual anchor positions are no longer estimated as independent states. Instead, they are functions of the anchor's position $\boldsymbol{r}_{\mathrm{a}}$ and the known floorplan. Let $f_i(\cdot)$ denote the mirroring transformation across the i-th reflecting surface (wall). The position of the i-th virtual anchor can be computed as $\boldsymbol{r}_{\mathrm{va}_i} = f_i(\boldsymbol{r}_{\mathrm{a}}(t_k))$. Consequently, the map-assisted observation vector $\boldsymbol{z}_{\mathrm{M}}(t_k) = \boldsymbol{h}_{\mathrm{M}}(\boldsymbol{x}_{\mathrm{M}}(t_k)) + \boldsymbol{v}_{\mathrm{M}}(t_k)$ directly maps the reduced state $\boldsymbol{x}_{\mathrm{M}}$ to the measurements:

$$\begin{aligned} z_{\mathrm{M}}(t_k) &= h_{\mathrm{M}}(\boldsymbol{x}_{\mathrm{M}}(t_k)) + v_{\mathrm{M}}(t_k) \\ &= ||\boldsymbol{r}_{\mathrm{r}}(t_k) - f_i(\boldsymbol{r}_{\mathrm{a}}(t_k))||_2 + c \cdot \Delta t(t_k) + v_{\mathrm{M}}(t_k). \end{aligned} \tag{9}$$

The Jacobian matrix $\mathbf{H}_{\mathrm{M}}(t_k)$ is defined as:

$$\mathbf{H}_{\mathrm{M}}(t_k) \triangleq \left.\frac{\partial h_{\mathrm{M}}}{\partial \boldsymbol{x}_{\mathrm{M}}}\right|_{\hat{\boldsymbol{x}}_{\mathrm{M}}(t_k)} = [\mathbf{H}_{\mathrm{M,r}}(t_k), \mathbf{H}_{\mathrm{M,a}}(t_k)], \tag{10}$$

While the receiver sub-matrix $\mathbf{H}_{\mathrm{M,r}}(t_k)$ is identical to the mapless case $\mathbf{H}_{\mathrm{r}}(t_k)$, the anchor sub-matrix $\mathbf{H}_{\mathrm{M,a}}(t_k)$ is different. Because the virtual anchor positions are functions of the anchor. Applying the chain rule to the i-th reflected path observation ($i \geq 1$) with respect to the anchor's position $\boldsymbol{r}_{\mathrm{a}}$ yields:

$$\frac{\partial z_{\mathrm{M},i}}{\partial \boldsymbol{r}_{\mathrm{a}}} = \frac{\partial ||\boldsymbol{r}_{\mathrm{r}} - f_i(\boldsymbol{r}_{\mathrm{a}})||_2}{\partial f_i(\boldsymbol{r}_{\mathrm{a}})} \cdot \frac{\partial f_i(\boldsymbol{r}_{\mathrm{a}})}{\partial \boldsymbol{r}_{\mathrm{a}}} = -\hat{\boldsymbol{e}}^T(f_i(\boldsymbol{r}_{\mathrm{a}}))\mathbf{M}_n$$

where $\hat{\boldsymbol{e}}^T(f_i(\boldsymbol{r}_{\mathrm{a}}))$ is the unit vector from the i-th virtual anchor to the receiver, and $\mathbf{M}_i \triangleq \partial f_i(\boldsymbol{r}_{\mathrm{a}})/\partial \boldsymbol{r}_{\mathrm{a}}$ is the Jacobian of the mirroring function. For a standard 2D reflection across a planar wall with a unit normal vector $\boldsymbol{n}_n$, this constitutes a Householder transformation matrix: $\mathbf{M}_i = \mathbf{I}_{2\times 2} - 2\boldsymbol{n}_i\boldsymbol{n}_i^T$. Consequently, the complete anchor sub-matrix is formulated as:

$$\mathbf{H}_{\mathrm{M,a}}(t_k) = \begin{bmatrix} -\hat{\boldsymbol{e}}^T(\boldsymbol{r}_{\mathrm{a}}) & -c & 0 \\ -\hat{\boldsymbol{e}}^T(f_1(\boldsymbol{r}_{\mathrm{a}}))\mathbf{M}_1 & -c & 0 \\ \vdots & \vdots & \vdots \\ -\hat{\boldsymbol{e}}^T(f_N(\boldsymbol{r}_{\mathrm{a}}))\mathbf{M}_i & -c & 0 \end{bmatrix}$$

Unlike the $\mathbf{H}_{\mathrm{a}}(t_k)$ in the mapless model where the lower block of the anchor's position Jacobian is entirely zero, the map-assisted model fills these entries with geometric reflection constraints ($-\hat{\boldsymbol{e}}^T(f_i(\boldsymbol{r}_{\mathrm{a}}))\mathbf{M}_i$). This indicates that every reflected multipath acts as an additional virtual ranging measurement, enhancing the observability and Fisher information of the anchor's state.

IV. OBSERVABILITY ANALYSIS

A. mapless model

We analyze the observability of different steps l based on the $\boldsymbol{\mathcal{O}}(t_0, t_l)$. For the mapless model. At $l = 1$, the system is severely unobservable. Let $\boldsymbol{\mathcal{O}}_u$ denote the u-th column vector of $\boldsymbol{\mathcal{O}}_1$. First, the columns corresponding to receiver velocity and all clock drifts are entirely zero (e.g., $\boldsymbol{\mathcal{O}}_3 = \boldsymbol{\mathcal{O}}_4 = \boldsymbol{\mathcal{O}}_6 = \boldsymbol{\mathcal{O}}_{10} = \mathbf{0}$), they are unobservable from a single snapshot. Second, the relative nature of the distance measurements creates linear dependencies among the remaining non-zero columns. For example, the receiver and anchor absolute clock bias columns satisfy $\boldsymbol{\mathcal{O}}_5 = -\boldsymbol{\mathcal{O}}_9$, indicating that only the relative clock bias is observable. Furthermore, the receiver's position columns ($\boldsymbol{\mathcal{O}}_{1:2}$) are linear combinations of the $2(N+1)$ columns corresponding to the positions of the anchor and all N virtual anchors. This structural column dependence mathematically proves that absolute time and global translation are fundamentally unobservable at $l = 1$.

For any $l \geq 2$, $\mathbf{F}$ activate the previously zero velocity ($\boldsymbol{\mathcal{O}}_{3:4}$) and clock drift ($\boldsymbol{\mathcal{O}}_6$) columns. However, four structural dependencies remain identical across all $l \geq 2$, defining an invariant null space: $\boldsymbol{\mathcal{O}}_5 + \boldsymbol{\mathcal{O}}_9 = \mathbf{0}$ (absolute clock bias), $\boldsymbol{\mathcal{O}}_6 + \boldsymbol{\mathcal{O}}_{10} = \mathbf{0}$ (absolute clock drift), and $\boldsymbol{\mathcal{O}}_{1:2} + \boldsymbol{\mathcal{O}}_{7:8} + \sum_{u=1}^{N} \boldsymbol{\mathcal{O}}_{10:10+2u} = \mathbf{0}$ (global 2D translation). Additionally, a fifth state-dependent dependency exists due to the global rotational gauge freedom. The positional and velocity columns are coupled through a weighted combination: $\sum_{u=1}^{4} \alpha_u \boldsymbol{\mathcal{O}}_u = \mathbf{0}$, where $\alpha_1 \triangleq (-y_r(t_0) + y_a(t_0))/\dot{x}_r(t_0), \alpha_2 \triangleq (x_r(t_0) - x_a(t_0)/\dot{x}_r(t_0), \alpha_3 \triangleq (-\dot{y}_r(t_0))/\dot{x}_r(t_0), \alpha_4 \triangleq 1$, ensuring that the range measurements remain invariant under any infinitesimal rotation of the entire trajectory around the anchor. Consequently, given the state-space dimension of $2N + 10$, the maximum achievable rank is bounded at $2N + 5$. At $l = 2$, the observability matrix $\boldsymbol{\mathcal{O}}_2$ provides only $2N + 2$ rows, which is strictly insufficient to reach the target rank. Only when $l \geq 3$ (with $N \geq 2$ multipaths) does the row dimension $(3N+3)$ satisfy the $2N+5$ requirement. This proves that $l = 3$ is the minimum temporal threshold to provide sufficient geometric constraints to decouple the receiver's velocity and clock drift from the static landmark geometry in a mapless system. The null space of $\boldsymbol{\mathcal{O}}(t_0, t_l)$ for $l \geq 3$ can be shown to be:

$$\mathcal{N}[\boldsymbol{\mathcal{O}}(t_0, t_l)] = \mathrm{span}[\boldsymbol{m}_1 \quad \boldsymbol{m}_2 \quad \boldsymbol{m}_3 \quad \boldsymbol{m}_4 \quad \boldsymbol{m}_5],$$

$$\boldsymbol{m}_1 = \boldsymbol{e}_5 + \boldsymbol{e}_9, \boldsymbol{m}_2 = \boldsymbol{e}_6 + \boldsymbol{e}_{10},$$

$$\boldsymbol{m}_3 = \boldsymbol{e}_1 + \boldsymbol{e}_7 + \sum_{u=1}^{N} \boldsymbol{e}_{11+2(u-1)}, \boldsymbol{m}_4 = \boldsymbol{e}_2 + \boldsymbol{e}_8 + \sum_{u=1}^{N} \boldsymbol{e}_{12+2(u-1)}$$

$$\boldsymbol{m}_5 = \alpha_1 \boldsymbol{e}_1 + \alpha_2 \boldsymbol{e}_2 + \alpha_3 \boldsymbol{e}_3 + \alpha_4 \boldsymbol{e}_4.$$

TABLE I
SYSTEM OBSERVABILITY COMPARISON: METRICS VS. INITIAL PRIORS FOR MAPLESS AND MAP-ASSISTED MODELS ($N \geq 3$)

Initial Prior (at t_0)	**Mapless Model ($2N+10$ States)**			**Map-assisted Model (10 States)**		
	Steps (l)	**Rank**	**Unobservable DoF**	**Steps (l)**	**Rank**	**Unobservable DoF**
None	$l \geq 3$	$2N+5$	Global Translation (2) Global Rotation (1) Clock Bias (1) Clock Drift (1)	$l \geq 2$	8	Clock Bias (1) Clock Drift (1)
Position	$l \geq 3$	$2N+7$	Global Rotation (1) Clock Bias (1) Clock Drift (1)	$l \geq 2$	8	Clock Bias (1) Clock Drift (1)
Time	$l \geq 3$	$2N+7$	Global Translation (2) Global Rotation (1)	$l \geq 2$	10	None
Position + Time	$l \geq 3$	$2N+9$	Global Rotation (1)	$l \geq 2$	10	None (Full Rank)

$\boldsymbol{e}_u$ is the standard basis vector consisting of a 1 in the u th element and zeros elsewhere.

B. map-aided model

Incorporating a prior floorplan reduces the state dimension to 10 ($\boldsymbol{x}_{\mathrm{M}}$) and enhances the system's observability. At $l = 1$, analyzing the columns of $\boldsymbol{\mathcal{O}}_1$ reveals a 5-dimensional null space due to zero columns for velocity ($\boldsymbol{\mathcal{O}}_{3:4}$) and clock drifts ($\boldsymbol{\mathcal{O}}_6, \boldsymbol{\mathcal{O}}_{10}$), alongside the strict linear dependence of clock biases ($\boldsymbol{\mathcal{O}}_5 + \boldsymbol{\mathcal{O}}_9 = \mathbf{0}$). However, the Householder transformation $\mathbf{M}_n$ embedded in the anchor's Jacobian ensures that the sum of the position columns is non-zero ($\boldsymbol{\mathcal{O}}_{1:2} + \boldsymbol{\mathcal{O}}_{7:8} \neq \mathbf{0}$). This proves that the map acts as an absolute spatial anchor, completely breaking the global 2D translation invariance even from a single snapshot. When extending to $l \geq 2$, $\mathbf{F}_{\mathrm{M}}$ activates the velocity and clock drift columns. Since the floorplan provides spatial but no temporal references, exactly two structural dependencies remain indestructible across all epochs: $\boldsymbol{\mathcal{O}}_5 + \boldsymbol{\mathcal{O}}_9 = \mathbf{0}$ (absolute clock bias) and $\boldsymbol{\mathcal{O}}_6 + \boldsymbol{\mathcal{O}}_{10} = \mathbf{0}$ (absolute clock drift), strictly bounding the maximum theoretical rank at $10 - 2 = 8$. Crucially, at $l = 2$, the observability matrix accumulates $2N+2$ rows. Assuming $N \geq 3$ multipaths, this instantly satisfies the rank requirement ($2N + 2 \geq 8$). Thus, for $N \geq 3$, unlike the mapless system which requires $l \geq 3$, the map-assisted model successfully resolves all absolute coordinates and relative states in just two epochs. The null space of $\boldsymbol{\mathcal{O}}(t_0, t_l)$ for $l \geq 2$ can be shown to be:

$$\mathcal{N}[\boldsymbol{\mathcal{O}}_{\mathrm{M}}(t_0, t_l)] = \mathrm{span}[\boldsymbol{m}_1 \quad \boldsymbol{m}_2],$$
$$\boldsymbol{m}_1 = \boldsymbol{e}_5 + \boldsymbol{e}_9, \ , \boldsymbol{m}_2 = \boldsymbol{e}_6 + \boldsymbol{e}_{10}.$$

Table I provides a comparative summary of how different prior influences the system's rank and Degrees of Freedom (DoF). The mapless model suffers from a persistent rotational freedom, even when the receiver's initial position and clock states are provided. In contrast, the map-assisted model leverages floorplan geometry to resolve all spatial ambiguities, requiring only a temporal reference to achieve full-state observability.

V. EVALUATION

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